Jie-En (Matthew) Yao

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Education

University of Southern California Aug 2023 – Expected May 2028
Doctor of Philosophy in Computer Science; GPA: 3.95/4.00 Los Angeles, CA

National Tsing Hua University Sep 2018 – Jun 2022
Bachelor of Science in Computer Science; GPA: 4.27/4.30; Class Rank: 1/142 Hsinchu, Taiwan

Research Experience

University of Southern California Aug 2023 – Present
Ph.D. Researcher, advised by Prof. C.-C. Jay Kuo Los Angeles, CA

- **Multimodal Database Index (under review):** Developed a multimodal index that augments image-only databases with textual descriptions for zero-shot composed image retrieval
- **Transductive Statistic Calibration (under review):** Developed a transductive method for few-shot class-incremental learning
- **HCL-FF (CVPR 2026):** Developed a hierarchical contrastive learning framework for Forward-Forward algorithm
- **Descrip3D (WACV 2026):** Improved LLM-based 3D scene understanding with object-level text descriptions

China Medical University Hospital Jan 2025 – Jan 2026
Research Assistant Los Angeles, CA (Remote collaboration)

- Built a coronary angiography pipeline using 619 angiograms from 462 patients, combining vessel segmentation (0.863 validation Dice), Fast Marching centerline extraction, and narrowed-to-normal vessel width ratios to quantify stenosis
- Developed an interactive clinical interface that enabled cardiologists to select coronary vessel segments and receive stenosis estimates and visualizations in under one second

MediaTek, Intelligence Vision Processing Team Jan 2022 – Dec 2022
AI Research Intern Hsinchu, Taiwan

- Designed Local Implicit Normalizing Flow (LINF), a generative method for arbitrary-scale image super-resolution that uses normalizing flows to model local texture distributions
- Built PyTorch training and evaluation pipelines across standard super-resolution benchmarks, achieving state-of-the-art perceptual quality across arbitrary scaling factors and a first-author publication at CVPR 2023

Selected Publications

Jie-En Yao, et al., *MDI: Multimodal Database Index for Zero-Shot Composed Image Retrieval*, under review, 2026
Jie-En Yao, et al., *Mahalanobis Classifier for Transductive Few-Shot Class-Incremental Learning*, under review, 2026
Jie-En Yao, et al., *HCL-FF: Hierarchical and Contrastive Learning for Forward-Forward Algorithm*, CVPR, 2026
Jintang Xue, **Jie-En Yao**, et al., *Descrip3D: Enhancing Large Language Model-based 3D Scene Understanding with Object-Level Text Descriptions*, WACV, 2026
Jie-En Yao, et al., *Local Implicit Normalizing Flow for Arbitrary Scale Image Super-Resolution*, CVPR, 2023

Work Experience

Industrial Technology Research Institute Jul 2020 – Sep 2020
Data Science Intern, Computational Intelligence Technology Center Hsinchu, Taiwan

- Built a quality prediction pipeline from 10,000+ raw manufacturing log records with 800+ variables, engineering features and training models with XGBoost and AutoGluon to reduce MAE by approximately 12% versus baseline

Skills

Programming and Machine Learning: Python, C++, PyTorch, NumPy, scikit-learn, Git
Research Areas: Computer Vision, Representation Learning, Continual Learning, Multimodal LLMs, Multimodal Retrieval, Medical AI
ML Systems: vLLM, Distributed Training, Mixed Precision, Model Profiling and Optimization, Linux, Docker